VIDEO GAME CREATION

Lesson 5 - Programming Basics IV: Enemy Ranks

Semester 1, Week 5

- **Objectives:** Explain what a loop is and why it avoids repeating code by hand; explain why well-structured per-instance behavior from Lesson 4 automatically scales to many instances without changes; explain Scratch's specific clone-identity limitation and its correct workaround; turn a single enemy into a row of independently-destructible enemies.
- **IGDA Tie-In:** Programming - loops/iteration, code reuse and object-oriented thinking (introductory)
- **Featured Engine (lecture demo):** GameMaker (GML Visual primary, with a brief GML script contrast - see this lesson's Sidebars)
- **Materials:** Demo laptop with all 8 engines verified (see Teacher's Checklist); Student Handouts, one per engine; students' own Lesson 4 projects (already saved from last week) - no new assets needed this week

Agenda

0:00-0:05	Welcome back, Lesson 4 recap, quick share of any working combat mechanics
0:05-0:25	Loops: Doing Something Many Times Without Repeating Yourself (live demo in GameMaker)
0:25-0:45	Scaling Behavior: Why One Enemy's Code Already Works for Many
0:45-0:50	A Scratch-Specific Wrinkle: Clones Have No ID
0:50-1:00	Introduce lab exercise, hand out engine-specific handouts
1:00-1:50	LAB: Turn your single enemy into a row of enemies in your ongoing [YourName]Game project
1:50-2:00	Exit check / quick share-out

Levels for this lesson's exercise

- Level 1 (Minimal): Several enemies appear in a row, created by a loop; shooting any one destroys just that one.
- Level 2 (Complete): Every enemy can be independently shot and destroyed with no errors; misses still behave exactly like last week.
- Level 3 (Refined): Sound already works for free; add row-to-row visual variety and/or confirm clean spawn-vs-behavior structure.

See: Lesson 5 Teacher's Script, Lesson 5 Prep Checklist, and the per-engine Student Handouts.